SOURCE CODE

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Import required libraries
import pandas as pd
from sklearn.metrics.pairwise import cosine_similarity
from sklearn.preprocessing import LabelEncoder
from google.colab import files

Upload CSV file
uploaded = files.upload()
filename = list(uploaded.keys())[0]

Load dataset
df = pd.read_csv(filename)

Encode movie names for matrix creation
movie_encoder = LabelEncoder()
df['movie_encoded'] = movie_encoder.fit_transform(df['movie'])

Create user-item matrix
user_item_matrix = df.pivot_table(index='userId', columns='movie_encoded', values='rating').fillna(0)

Compute similarity between users
user_similarity = cosine_similarity(user_item_matrix)

Convert to DataFrame
similarity_df = pd.DataFrame(user_similarity, index=user_item_matrix.index,
columns=user_item_matrix.index)

def get_recommendations(user_id, top_n=3):
    Get similarity scores for all users except the target user
    similar_users = similarity_df[user_id].drop(user_id).sort_values(ascending=False)

    Filter the user-item matrix to include only similar users
    similar_users_matrix = user_item_matrix.loc[similar_users.index]

    Calculate weighted scores using the filtered matrix
    weighted_scores = similar_users_matrix.T.dot(similar_users) / similar_users.sum()

    user_rated_movies = user_item_matrix.loc[user_id]
    recommendations = weighted_scores[user_rated_movies ==
0].sort_values(ascending=False).head(top_n)
    recommended_movie_ids = recommendations.index
    recommended_movies = movie_encoder.inverse_transform(recommended_movie_ids)
    return list(recommended_movies)
```